

## JAVASCRIPT FUNDAMENTALS

## FORMATIVE ASSESSMENT

## KM-01: Introduction to JavaScript Programming

## ELECTRONIC LEARNER ASSESSMENT PAPER

|                     |                                          |
|---------------------|------------------------------------------|
| Skills Programme    | SP-220373: JavaScript Programmer         |
| Module code         | 900219-000-00-KM-01                      |
| NQF level / credits | Level 4 / 6 credits                      |
| Assessment method   | Electronic internal formative assessment |
| Duration            | 120 minutes                              |
| Assessment decision | Competent (C) / Not Yet Competent (NYC)  |

## Learner details

|                     |                                       |
|---------------------|---------------------------------------|
| Full name           | Shane Davis                           |
| ID / learner number | 8309175131082                         |
| Cohort / group      | Group 1                               |
| Assessment date     | 21 July 2026                          |
| Electronic filename | Davis_Shane_KM01_Formative_Assessment |

## How to complete and submit this assessment

- Download this file from the LMS and open it in Microsoft Word. Select Enable Editing if Word displays a protected-view message.
- Complete the yellow fields. Use each dropdown to select an answer for multiple-choice, True/False and matching questions.
- Click inside each written-response field and type your answer. The field will expand as you type.
- Answer every question independently. Do not use the internet, AI tools or another person's assistance unless an approved accommodation applies.
- Save regularly. Before submission, rename the file using:  
Surname\_Name\_KM01\_Formative\_Assessment.docx.
- Upload the completed Word document to the correct LMS submission link. Do not convert it to PDF.

**Competency-based assessment**

No numerical mark or percentage is awarded. The assessor reviews the complete evidence against the KM-01 Internal Assessment Criteria and records either Competent (C) or Not Yet Competent (NYC). An NYC decision is followed by feedback, remediation and another opportunity to demonstrate the outstanding competence.

## Assessment structure

| Question   | Assessment method          | Electronic response method                       |
|------------|----------------------------|--------------------------------------------------|
| Question A | Multiple-choice            | Select one option from each dropdown.            |
| Question B | True or False              | Select T or F from each dropdown.                |
| Question C | Mix and Match              | Select the matching letter from each dropdown.   |
| Question D | Written responses          | Type a clear response into each expanding field. |
| Question E | Integrated retail scenario | Apply KM-01 knowledge in typed responses.        |

### SECTION A | Multiple-choice questions

#### ELECTRONIC RESPONSE

Select the best answer from the dropdown below each question. Choose one answer only.

#### 1. Which item is system software that manages hardware and allows applications to run?

- A. A keyboard
- B. An operating system
- C. A web page
- D. A folder

**Your answer:** B

#### 2. Which activity specifically depends on internet connectivity?

- A. Renaming an offline file
- B. Using a cloud-storage service
- C. Turning on a monitor
- D. Creating a local folder

**Your answer:** B

#### 3. Which statement best defines JavaScript?

- A. A markup language for page structure
- B. A stylesheet language for colours and layout
- C. A scripting/programming language used to add behaviour
- D. A database used to store records

**Your answer:** C

#### 4. Client-side JavaScript normally runs...

- A. inside the user's web browser
- B. only inside a database
- C. only on a printer
- D. inside a CSS file

**Your answer:** A

#### 5. Which line is a single-line JavaScript comment?

- A. <!-- note -->
- B. // note
- C. /\*/ note
- D. # note

**Your answer:** B

#### 6. Which statement correctly identifies a limitation or caution?

- A. JavaScript automatically secures every website

- B. JavaScript removes the need for testing
- C. JavaScript can contain errors and must be tested
- D. JavaScript replaces both HTML and CSS

**Your answer:** C

**7. Which is a recognised application of JavaScript?**

- A. Web applications
- B. Physical cable manufacture
- C. Paper filing only
- D. Mechanical furniture assembly only

**Your answer:** A

**8. A furniture store wants a page to calculate a promotional discount immediately. Which technology should perform the calculation?**

- A. HTML
- B. CSS
- C. JavaScript
- D. A JPEG image

**Your answer:** C

**9. Which IDE/editor feature helps a learner find a JavaScript error?**

- A. Debugging tools
- B. Wallpaper settings
- C. Printer toner control
- D. Screen brightness

**Your answer:** A

**10. What does refactoring mean?**

- A. Changing code structure without changing required behaviour
- B. Deleting the whole repository
- C. Publishing without testing
- D. Replacing JavaScript with an image

**Your answer:** A

**11. Which statement correctly distinguishes Git from GitHub?**

- A. Git tracks changes; GitHub hosts and supports collaboration
- B. Git is a browser; GitHub is an operating system
- C. Git styles pages; GitHub structures pages
- D. They are unrelated to version control

**Your answer:** A

**12. What is the purpose of a pull request?**

- A. To print the page
- B. To ask for review before merging changes
- C. To delete all commits
- D. To refresh a browser tab

**Your answer:** B

**13. Breaking “build a support form” into smaller tasks is called...**

- A. decomposition
- B. redirection
- C. deployment
- D. compression

**Your answer:** A

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**14. Which is an edge case for a required order-number field?**

- A. A valid order number is entered
- B. The field is left blank or contains spaces only
- C. The page heading is visible
- D. The CSS file loads

**Your answer:** B

**15. At which life-cycle stage should expected and actual results be compared?**

- A. Testing
- B. Design only
- C. File naming only
- D. Release without review

**Your answer:** A

**16. Which item belongs in a beginner Definition of Done?**

- A. The feature works and has been tested
- B. The files remain unsaved
- C. Console errors are ignored
- D. No evidence is retained

**Your answer:** A

**17. What is a cookie?**

- A. A browser-associated piece of stored data
- B. A physical hardware component
- C. A JavaScript IDE
- D. A Git branch

**Your answer:** A

**18. Which is responsible cookie practice?**

- A. Store passwords in readable cookies
- B. Ignore expiry dates
- C. Use cookies carefully and protect session data
- D. Share session cookies with colleagues

**Your answer:** C

**19. Which JavaScript command opens the browser print dialog?**

- A. window.print()
- B. window.pull()
- C. console.branch()
- D. document.commit()

**Your answer:** A

**20. Which object can be used to redirect the browser to another page?**

- A. window.location
- B. console.log
- C. style.css
- D. git.commit

**Your answer:** A

## SECTION B | True or False

ELECTRONIC  
RESPONSE

Select T for True or F for False in the answer field for each statement.

| No. | Statement                                                                                 | Electronic answer |
|-----|-------------------------------------------------------------------------------------------|-------------------|
| 1   | A processor and operating system contribute to the platform on which software runs.       | T — True          |
| 2   | JavaScript and Java are the same programming language.                                    | F — False         |
| 3   | JavaScript can run both client-side and server-side when a suitable runtime is available. | T — True          |
| 4   | JavaScript is used only for pop-up messages and has no other applications.                | F — False         |
| 5   | An IDE may provide code completion, formatting, refactoring and debugging support.        | T — True          |
| 6   | A commit is a meaningful saved point in a repository's history.                           | T — True          |
| 7   | Problem-solving should begin by coding immediately without clarifying the requirement.    | F — False         |
| 8   | Testing and ongoing support are parts of a solution-development life cycle.               | T — True          |
| 9   | Cookies should be used to store sensitive information without security controls.          | F — False         |
| 10  | A page refresh reloads a page, while a redirect sends the browser to a different address. | T — True          |

## SECTION C | Mix and Match

ELECTRONIC  
RESPONSE

Match each concept in Column A to the correct explanation in Column B. Select the correct letter in the electronic answer column.

| Column A: Concept             | Electronic answer | Column B: Explanation options                                               |
|-------------------------------|-------------------|-----------------------------------------------------------------------------|
| 1. Platform                   | F                 | A. A proposal asking others to review changes before merging.               |
| 2. Client-side                | B                 | B. Code execution inside the user's browser.                                |
| 3. <code>console.log()</code> | E                 | C. Checking actual behaviour against expected results.                      |
| 4. Server application         | I                 | D. A rule controlling when stored cookie data should stop being valid.      |
| 5. Refactoring                | J                 | E. Built-in browser-console function used for messages and basic debugging. |
| 6. Pull request               | A                 | F. Processor/operating-system environment in which software runs.           |
| 7. Decomposition              | H                 | G. Browser object used to read or change the page address.                  |
| 8. Testing                    | C                 | H. Breaking a large problem into smaller parts.                             |

| Column A: Concept   | Electronic answer | Column B: Explanation options                                    |
|---------------------|-------------------|------------------------------------------------------------------|
| 9. Cookie expiry    | D                 | I. Software that runs on a server and responds to requests.      |
| 10. window.location | G                 | J. Improving code structure without changing intended behaviour. |

**SECTION D | Written responses****TYPE YOUR RESPONSE**

Type a clear response in each yellow field. The fields expand as you type.

**1. Explain how hardware, software, a processor and an operating system work together as a computer platform. Briefly explain 8-bit representation and give one internet-enabled or remote-working capability that a learner may use.**

The physical components of a computer system, such as the computer itself, keyboard, memory, and storage devices, are referred to as hardware.

Programs and apps that give instructions, enabling the hardware to carry out particular duties, are referred to as software.

The processor (CPU) manages data processing, computations, and program execution.

In addition to providing the platform on which programs operate, the operating system (OS) manages and synchronizes the computer's hardware, memory, files, and connected peripherals.

Eight binary digits (bits) consisting of 0s and 1s make constitute an 8-bit representation, sometimes referred to as one byte. It has an unsigned range of 0 to 255 and can hold 256 distinct values.

**2. Define JavaScript and explain where it runs. State one useful feature, one limitation or caution, one built-in function with its purpose, and the purpose of comments.**

JavaScript is a computer language that gives webpages more capability and engagement. It mostly operates within a user's web browser, but when utilized with a runtime like Node.js, it can also operate on a server.

JavaScript's ability to rapidly update web page content without requiring a page reload is one of its useful features. One drawback is that JavaScript code may contain flaws, thus it must be carefully checked to ensure proper operation.

`Console.log()` is one built-in function that shows messages in the browser console for debugging and testing.

To make the code easier to comprehend, maintain, and collaborate on with other developers, comments are used to describe what the code does.

**3. Define an IDE or code editor and state its purpose. Describe two useful features. Then distinguish between refactoring and debugging.**

Software that makes it easier for developers to write, modify, and manage code is called an IDE (Integrated Development Environment) or code editor. Its primary goal is to simplify the coding process by offering practical tools in one location.

Code completion, which makes code suggestions as you type to save time and minimize errors, and debugging tools, which assist you in identifying and correcting software issues, are two useful features.

Refactoring is the process of making your code more readable and structured without altering its functionality. In contrast, debugging is the process of identifying and correcting flaws so that the software operates as intended.

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**4. Distinguish Git from GitHub. Explain the purpose of a repository, branch, commit and pull request in a team workflow.**

Git is a version control system that keeps track of changes made to files, while GitHub is an online platform that stores Git repositories and allows developers to work together on projects.

A repository is where all the project files and their version history are stored. A branch lets developers work on new features or fixes without affecting the main version of the project. A commit is a saved snapshot of changes made to the project, usually with a message describing what was updated. A pull request is used to ask team members to review the changes before they are merged into the main branch.

**5. Define a cookie and state one legitimate purpose. Explain, at awareness level, what it means to store, read, set an expiry date for and delete a cookie. Include one security or privacy caution.**

A cookie is a tiny bit of information that a website saves in your browser. Keeping a person logged in or remembering their preferences when they return to the website is a typical and acceptable application for cookies.

Information is saved in the browser when a cookie is stored. When a cookie is read, the website can access the data that has been stored. The amount of time the cookie stays on the device before being automatically deleted is determined by setting an expiration date. When a cookie is deleted, it is removed from the browser and is no longer functional.

Sensitive data, including passwords and personal information, should never be saved in cookies without appropriate security safeguards. This is a crucial security and privacy precaution.



**SECTION E | Integrated retail scenario****TYPE YOUR RESPONSE****Scenario: Furniture delivery enquiry page****Workplace brief**

A Lewis Group furniture store wants a simple internal browser page that sales consultants can use to check a customer delivery enquiry. The page must display a heading and instructions, accept an order number, ask the consultant to confirm the enquiry, show a clear message, allow a summary to be printed, and redirect to a follow-up page when required. The first version will not connect to a database. The team will use VS Code, test in a browser, inspect the console, and collaborate through GitHub.

**1. List four recognised application areas for JavaScript. Then identify two appropriate uses of JavaScript in this delivery-enquiry page.**

- Creating interactive websites.
- Developing web applications.
- Building server-side applications using environments such as Node.js.
- Creating browser-based games and interactive multimedia.

For the Lewis Group delivery-enquiry page, JavaScript can be used to:

Validate the order number entered by the sales consultant and display a confirmation or enquiry message.

Print a summary of the enquiry and redirect the browser to a follow-up page when needed.

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**2. Apply problem-solving before coding: break the page into three smaller tasks and identify one input, one process and one output.**

I would break the project up into smaller, easier-to-manage activities before I started coding:

Include instructions, a title, and a box for the consultant to enter the order number while designing the webpage.

Create the JavaScript code to verify the order number, request confirmation of the inquiry from the consultant, and show an appropriate message.

Provide the option to print an inquiry summary and, if needed, direct the user to the follow-up page.

Applying the Input-Process-Output (IPO) method:

Input: The customer's order number is entered by the consultant.

Procedure: The JavaScript verifies the data entered, requests confirmation, then handles the inquiry.

Output: The page displays a message with the choice to either print the inquiry details or proceed to the follow-up page.

**3. Describe six relevant stages or activities in the development life cycle, from understanding the requirement through release or ongoing support. Apply each one briefly to the scenario.**

The following is an explanation of the Lewis Group delivery-enquiry page's development life cycle:

1. Recognizing the requirements: The first stage is to determine what the page must accomplish, such as enabling consultants to enter an order number, verify the inquiry, print a summary, and go to a follow-up page.

2. Designing the solution: Consider the layout of the heading, directions, input field, buttons, and messages as well as how users will interact with the page.

3. Building the page: Create the solution using JavaScript to manage the functionality of the page, HTML for the webpage structure, and CSS for styling.

4. Testing the solution: Use a browser to test the page to make sure every feature functions as it should. Make that the order number is appropriately accepted, messages appear correctly, printing functions, and the user is redirected to the appropriate page.

5. Page release: Following successful testing, make the page accessible for the sales consultants to utilize on a daily basis.

6. Ongoing maintenance: After the page is made available, keep resolving problems, enhancing functionality, and updating it as new needs emerge.

**4. Identify the suitable JavaScript/browser feature for each need: collect an order number, request OK/Cancel confirmation, open printing, redirect to another page, and reload the current page. Finally, state the basic purpose of javascript:void(0) when used in a link.**

For the delivery-enquiry page, the following JavaScript and browser features are suitable for each task:

- Collect an order number: Use an input field to allow the consultant to enter the customer's order number.
- Request an OK/Cancel confirmation: Use the confirm() function to ask the consultant to confirm the enquiry before continuing.
- Open the print dialog: Use window.print() to allow the consultant to print a summary of the enquiry.
- Redirect to another page: Use window.location or window.location.href to send the user to the follow-up page.
- Reload the current page: Use window.location.reload() to refresh the current page.

The main purpose of javascript:void(0) in a link is to stop the browser from navigating to another page when the link is clicked. It is often used when the link is meant to trigger JavaScript code instead of opening a new page.

**5. Explain the separate roles of HTML, CSS and JavaScript in this page. State why the JavaScript is client-side, give one suitable code comment, and state one limitation or professional caution about browser dialogs.**

The headline, directions, input box, and buttons on this delivery-enquiry page are all created using HTML. The page's look, including its colors, fonts, spacing, and general layout, is controlled by CSS. The interactive elements, such as verifying the order number, printing the inquiry summary, showing confirmation messages, and rerouting the user to another page, are added via JavaScript.

Because JavaScript operates directly in the user's web browser, it is client-side and enables instantaneous action without sending each request to a server.

A suitable code comment could be:

```
// Check that the order number has been entered before continuing
```

One expert warning regarding browser dialogs is that they should not be overused because numerous pop-up notifications might disrupt the user's workflow and lower the user experience in general.

## Learner declaration and electronic submission check

### Learner declaration

I declare that the responses in this document are my own work, except for any approved accommodation recorded by the assessor. I understand that copied, AI-generated or unauthorised responses may be handled according to the provider's assessment and academic-integrity policies.

|                           |                                       |
|---------------------------|---------------------------------------|
| Learner's typed full name | Shane Davis                           |
| Date completed            | 24 July 2026                          |
| Final saved filename      | Davis_Shane_KM01_Formative_Assessment |

### Assessor decision and feedback

|                       |                           |
|-----------------------|---------------------------|
| Overall decision      | Select decision ▼         |
| Assessor name         | Click or tap here to type |
| Decision date         | Click or tap here to type |
| Reassessment required | Select decision ▼         |

### Feedback: demonstrated competence and outstanding gaps

Click or tap here to type your response. The field will expand as you type.

### Remediation action and reassessment date

Click or tap here to type your response. The field will expand as you type.